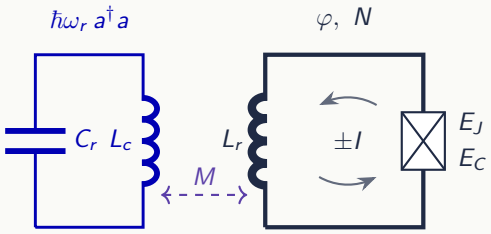
A π loop coupled to a quantum LC circuit: spontaneous persistent current, exactly

From the rf-SQUID circuit model to a quantum-dot Josephson loop solved by DMRG

Group seminar

The circuit

The circuit: an rf-SQUID and a detached LC circuit



- **Loop:** a Josephson junction (E_J, charging energy E_C) closing an inductance L_r (E_L). Biased at half a flux quantum, $\varphi_{\text{ext}} = \pi -$ or closed by an intrinsic π junction – it is a π loop.
- LC **circuit**, galvanically detached, mode a at ω_r: its flux $MI_c \propto a + a^\dagger$ threads the loop.
- No current drive, only a seed $\varphi_{\text{ext}} = \pi - 10^{-3}$ to pick a direction.

Question: does the quantum flux of the LC circuit make the loop carry a persistent current?

The two-mode Hamiltonian (original notebook)

$$H = E_C\, N^2 + E_L\big(\varphi - \varphi_{\rm ext} - g\,(a + a^\dagger)\big)^2 - E_J \cos\varphi + \hbar\omega_r\, a^\dagger a, \qquad \varphi = \tfrac{1}{\sqrt{2}}\big(\tfrac{E_C}{E_L}\big)^{1/4}(b + b^\dagger), \;\; [\varphi, N] = i$$

Resonator seen statically (minimize over $a + a^\dagger$)

$\frac{1}{E_L^{\rm eff}} = \frac{1}{E_L} + \frac{4g^2}{\hbar\omega_r}$: the LC circuit *adds inductance* to the loop (M^2/L_c). Screening parameter $\beta_{\rm eff} = \frac{E_J}{2E_L^{\rm eff}} = \frac{E_J}{2E_L}(1 + 4g_1^2)$, $g_1^2 \equiv g^2 E_L/\hbar\omega_r$: the only combination the static problem knows.

Notebook ($E_C = 1$): $E_L = 0.5$, $E_J = 2$: $\beta_{\rm loop} = 2$, but the bare well ($0.84 E_C$) lies below $\omega_{rf} = 2\sqrt{E_C E_L} = 1.4$ – no polarization without the resonator.

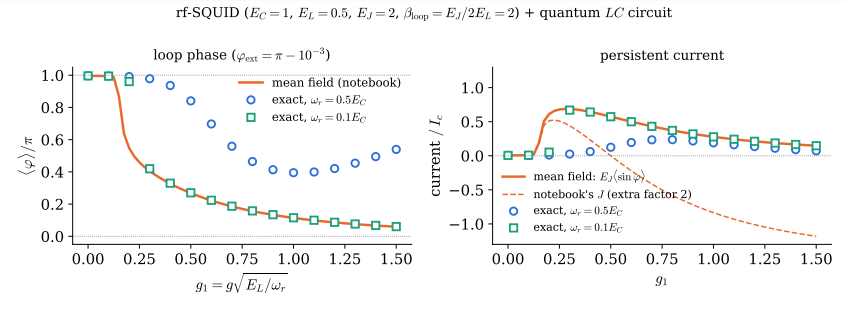
Mean field of the notebook: $|{\rm GS}\rangle = |\psi_{rf}\rangle|\psi_{ph}\rangle$

Resonator in a coherent state $\alpha \propto \langle\varphi\rangle$, SQUID mode b exact (200 Fock states):

$$H_{rf}^{\rm mf} = E_C N^2 + E_L\varphi^2 - E_J\cos\varphi - \frac{2E_L g_1^2}{\frac{1}{4} + g_1^2}\,\langle\varphi\rangle\,\varphi$$

($\varphi \to \varphi - \varphi_{\rm ext}$), self-consistent in $\langle\varphi\rangle$ = Hartree linearization of the exact static term. The resonator's zero-point motion is gone: mean field depends on g_1 only, the exact problem on g and ω_r.

What the notebook predicts, and what the exact two-mode solution says



Mean field (reproduced to all digits): the loop polarizes at $g_1 \simeq 0.15$–0.3, the current peaks at $0.69\,I_c$ and then falls as the loop lets in half a flux quantum ($\langle\varphi\rangle \to 0$). Caveat: the notebook evaluates J with $(1-2c)$ instead of $(1-c)$, $c = g_1^2/(\frac{1}{4}+g_1^2)$; the correct $-\partial E/\partial\varphi_{\rm ext}$ equals $E_J\langle\sin\varphi\rangle$ (solid).

Exact ground state of both modes (seed 10^{-3}): for a slow resonator ($\omega_r = 0.1E_C$) it follows the mean field; for $\omega_r = 0.5E_C$ the state stays a cat and the current is $\leq 0.24\,I_c$. Zero-point motion of the resonator opposes the polarization – exactly what the microscopic model will show.

The limit we then solve exactly: loop flux locked to the LC flux

Take $E_L \to \infty$ ($k = M/\sqrt{L_r L_c} \to 1$): $E_L(\varphi - g(a + a^\dagger))^2$ locks $\varphi = g(a + a^\dagger)$ and the loop mode b ($\omega_{rf} \to \infty$) freezes out:

$$H \;\to\; \hbar\omega\, a^\dagger a + E_J \cos\!\left[2g\,(a + a^\dagger)\right]$$

for a π junction ($+E_J \cos$; $2g$ = pair-phase coupling, $g^2 = \pi Z/4R_Q$). This is the circuit of the manuscript: the loop *is* the inductor of the mode (fluxonium-like, galvanic). In the pair phase $\theta = 2gX$:

$$H = 4E_C\, n^2 + \tfrac{1}{2} E_L \theta^2 + E_J \cos\theta, \quad E_L = \frac{\hbar\omega}{8g^2},\; E_C = \hbar\omega g^2,$$

$\beta = E_J/E_L = 8g^2 E_J/\hbar\omega = 2\pi L I_c/\Phi_0$: the $4g_1^2$ term of the two-mode model alone.

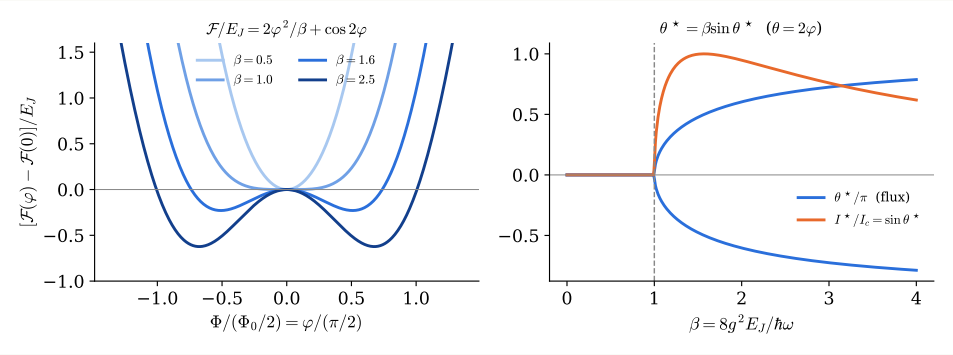
Why this limit

It keeps the mechanism (the resonator's inductance and zero-point flux act on the junction) with one bosonic mode, so that the junction can be replaced by the *microscopic* interacting ring and solved by DMRG.

What the inductive case ($k < 1$) adds

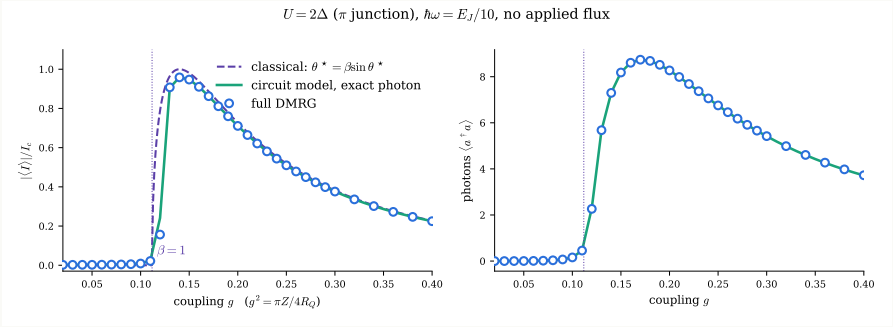
Same classical threshold ($L \to L_r^{\mathrm{total}}$) and cat physics, but two zero-point fluxes sharing δ, the photon numbers and the flux-driven 0–π shift: a second bosonic site in the DMRG, not yet computed.

Classical picture: a double well and a pitchfork



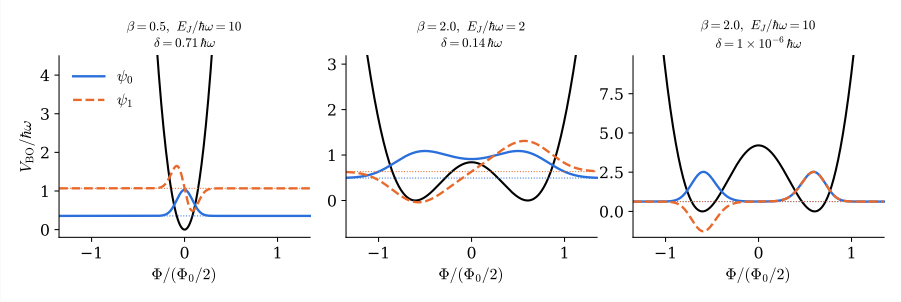
Minima at $\theta^\star = \beta \sin\theta^\star$ carry the current $I^\star = I_c \sin\theta^\star$ and a static flux $\langle\Phi\rangle = LI^\star$: a "photon condensate" $\langle a\rangle \neq 0$. The exact identity $\langle I\rangle = \langle\Phi\rangle/L$ holds in every quantum eigenstate.

The persistent current that emerges



- Above $g_c \simeq 0.13$: a current of $0.96\, I_c$ and a condensate of 7–9 photons. The threshold lies slightly above the classical $\beta = 1$; beyond it the current falls as $\hbar\omega/g^2$ (the flux saturates at $\Phi_0/2$).
- Circles: full microscopic solution (next part); line: circuit Hamiltonian with an exact photon – they agree to 0.2% of I_c.

What is the quantum ground state? A cat



Exact eigenstates of the circuit Hamiltonian. For $\beta > 1$ the ground state is the symmetric superposition $(|\circlearrowleft\rangle + |\circlearrowright\rangle)/\sqrt{2}$ of the two current directions, split from its partner by a tunnel gap $\delta \sim \hbar\omega\, e^{-(E_J/\hbar\omega)\, s(\beta)}$.

A tiny flux offset ($\delta\Phi > \delta/2I^\star$) selects one direction. If $E_J \lesssim \hbar\omega$ the well is shallower than the zero-point energy and no current state exists at all (middle panel).

Where does $E_J \cos\theta$ come from? The microscopic model

The microscopic model: a quantum dot in a BCS ring, plus the mode

$$H = \hbar\omega\, a^\dagger a \underbrace{- t \sum_{r,\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.}\right) + \Delta \sum_{r} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.}\right)}_{\text{BCS ring, } L-1 \text{ sites}} + \underbrace{\epsilon_d\, n_0 + U\, n_{0\uparrow} n_{0\downarrow}}_{\text{Anderson dot}}$$

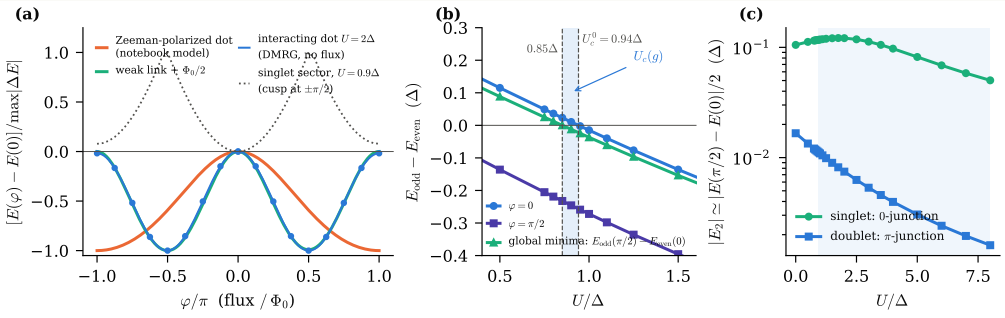
$$- t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.}\right) - t_J \sum_{\sigma} \left(e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.}\right)$$

- The flux is a Peierls phase on the dot bond; quantized, it becomes the displacement operator $e^{ig(a+a^\dagger)}$ (diamagnetic terms to all orders).
- Parity is conserved: **even** = singlet, a 0 junction; **odd** = Kramers doublet, a π junction (a Cooper pair crosses the singly occupied dot with its spins reversed).

Solved exactly by DMRG on the full electron⊗photon space ($\chi = 300$, Fock cutoff 50–60, sum rule $\langle I \rangle = \langle \Phi \rangle / L$ as acceptance test).

$\Delta = t$, $t_J = 0.5\Delta$, $\epsilon_d = -U/2$, $L = 10$, $\hbar\omega = E_J(2\Delta)/10$.

The junction alone: 0 or π is decided by U

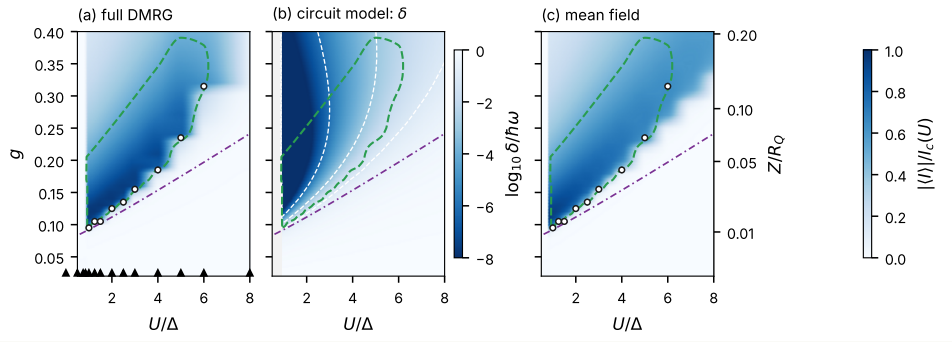


(a) Ring energy vs flux. The doublet at $U = 2\Delta$ is a pure π junction, $E_2 = 7.39 \times 10^{-3}\Delta$; at zero flux it looks exactly like an ordinary weak link biased by $\Phi_0/2$.

(b) Singlet–doublet crossing at $U_c^0 = 0.94\Delta$ (zero flux); the *global* minima cross already at 0.85Δ.
(c) E_J of the π junction falls as $U^{-1.3}$: $E_J/\hbar\omega$ runs from 15 to 2 across the phase diagram.

The phase diagram

Phase diagram in the (U, g) plane – full DMRG



(a) Full DMRG, 427 ground states, seed 10^{-3}: persistent current $|\langle I \rangle|/I_c(U)$. Triangles: computed U; circles: $0.5 I_c$.

(b) Seed-free order parameter: tunnel splitting δ of the circuit model with the DMRG $E(\varphi)$. Grey: singlet 0 loop.

(c) Mean field (product state). Green dashed: $0.5 I_c$ of the circuit model; purple: classical $\beta = 1$. Right axis: Z/R_Q.

How to read it

$U < U_c$: a 0 loop

Singlet ground state, diamagnetic: the mode is *stiffened*, no current. The boundary moves from 0.94Δ to 0.85Δ with g: the zero-point flux itself drives a $0 \to \pi$ switch.

$U > U_c$, $g > g_c(U)$: the condensate

Current $\simeq I_c$, 7–9 photons. g_c follows $\beta = 1$ but lies above it by $\Delta g = 0.02$, 0.04, 0.12 at $E_J/\hbar\omega = 10$, 5, 3.2: zero-point motion opposes the condensation.

$E_J \lesssim 2.5\,\hbar\omega$ ($U \gtrsim 8\Delta$): no order

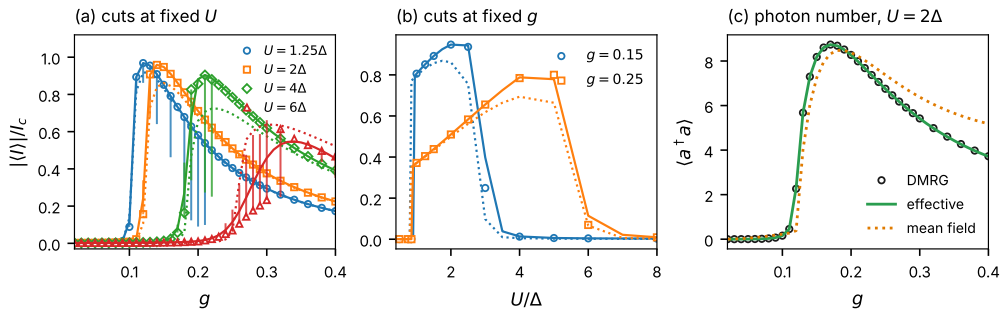
β reaches 2.7, yet the ground state is the cat, $\delta \simeq \hbar\omega$, current $\propto$ seed.

Mean field (c) fails where it matters

Orders at $U = 7$–8Δ with $0.6 I_c$ (absent) and at $U = 0.85\Delta$ with $0.95 I_c$ where the exact state is a 0 loop: a product state cannot be a cat, and replaces the Kramers doublet by a static moment.

Full DMRG = circuit Hamiltonian to 0.2% of I_c: here the loop *is* a π-fluxonium with the microscopic $E(\varphi)$.

Cuts through the diagram



Symbols: full DMRG; solid: circuit model; dotted: mean field. The sharp step at $U = 1.25,\ 2\Delta$ ($E_J/\hbar\omega \simeq 13,\ 10$) becomes a broad crossover at $U = 6\Delta$ ($E_J/\hbar\omega = 3.2$).

The mean-field approach of the ring notebook, and a caveat

The mean-field approach of the ring notebook

Same product-state idea, now for the microscopic ring (`Optimised_SC_ring_with_JJ_cavity_mean_field.nb`): one dot + 10 BCS sites ($t_0 = 2$, $\Delta_0 = 0.6$, $t_R = t_L = 0.5$), the dot *Zeeman-polarized* ($\epsilon_\uparrow = -0.3$, $\epsilon_\downarrow = 3.7$: a magnetic dot = Hartree–Fock image of a large-U dot), one mode $\hbar\omega = 0.02$, seed $\varphi_{\rm cl} = 10^{-3}$, $g = 0$–1.2.

electrons: BdG with the junction bond $t_R \to t_R\,\alpha, \quad \alpha = \langle\psi_{ph}|e^{igX}|\psi_{ph}\rangle$

photon: $\hbar\omega a^\dagger a - \left(\eta\, e^{igX} + \text{h.c.}\right), \quad \eta = t_R e^{i\varphi_{\rm cl}}\langle c_1^\dagger c_0\rangle$

$$E = E_{\rm el} + E_{\rm ph} + 2\,\mathrm{Re}(\eta\alpha), \qquad I = -2\,\mathrm{Im}(\eta\alpha)$$

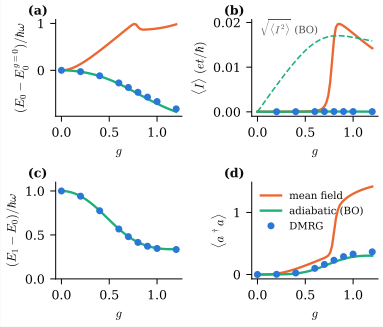
What it predicts

A current-carrying phase for $g \gtrsim 0.75$: $|I| \approx 0.02$, $\langle a^\dagger a\rangle \simeq 1$–$1.4$, $\langle X\rangle \neq 0$ – the "photon condensate" of the circuit model, now from a microscopic ring.

Checks

The Python port reproduces all five notebook curves to machine precision; the algebra (Nambu convention, double counting, $I = -\partial E/\partial\varphi$) is correct. The question is only the *ansatz*.

Caveat: with exactly these parameters, DMRG finds nothing – and why



Full DMRG (electrons $\otimes$ photon, same H, same seed): no current at any g, only a linear response 7×10^{-5}; exact energy $1.8\hbar\omega$ *below* mean field, gap stays $0.33\hbar\omega$, $\langle a^\dagger a \rangle \simeq 0.36$. The circuit model (exact photon) agrees with DMRG.

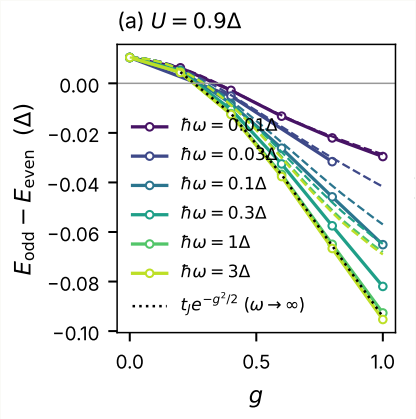
Three reasons

(1) $E_J \simeq \hbar\omega$: the double well ($g > 0.58$) is shallower than the zero-point energy; the ground state is a cat, which a product state cannot be – it *self-traps* in one well.
(2) The product state is not gauge covariant: it shrinks the junction bond by $|\langle e^{igX} \rangle| < 1$ and loses phase-independent kinetic energy that the exact state recovers by following the flux.
(3) $L = 11 \lesssim \xi$: $E(\varphi)$ is dominated by the h/e harmonic ($E_1 = 2.1 \times 10^{-2} \gg E_2 = 2 \times 10^{-3}$): the "instability" would be of the normal persistent current, not of the junction.

Same model at $\hbar\omega = 0.002$: it *does* order ($g_c = 0.21$; mean field 0.30). The control parameter is $E_J/\hbar\omega$ – hence the manuscript's $E_J/\hbar\omega = 10$ and $L \gg \xi$.

Outlook

Bonus: the vacuum of the mode switches the junction from 0 to π



$U = 0.9\Delta$: a 0 junction without the mode becomes a π junction at $g \simeq 0.3$ ($Z \simeq 0.6\,\mathrm{k\Omega}$).

- Both parity sectors by full DMRG, no seed, $\hbar\omega = 0.01$–$3\,\Delta$: U_c of the loop moves down by 12% at $g = 0.5$ and 30–40% at $g = 1$.
- Slow mode: zero-point flux costs the singlet energy (its surface curves up) and pays the doublet. Fast mode: the vacuum average $e^{-g^2/2}$ of the Peierls factor reduces the level width (Franck–Condon).
- Observable as the resonator-frequency jump of a gate-tuned dot junction, driven by the impedance alone.

Summary

Summary

1. A π junction sharing its inductance with an LC mode is a π-fluxonium, $H = 4E_C n^2 + \frac{1}{2}E_L\theta^2 + E_J\cos\theta$: for $\beta = 8g^2E_J/\hbar\omega > 1$ a persistent current and a photon condensate appear; the exact ground state is a cat of the two current directions.
2. The full-DMRG phase diagram of the microscopic model (dot + BCS ring + photon) confirms it without approximation: condensate for $U > U_c$, $g > g_c(U)$; threshold pushed above $\beta = 1$ by zero-point motion; no order for $E_J \lesssim 2.5\,\hbar\omega$; mean field fails exactly there and at $U \simeq U_c$.
3. The zero-point flux of the mode is itself a control parameter of the 0–π transition.

Thank you!

Backup: physical units

$$E_J/h \simeq 0.5\,\mathrm{GHz} \times I_c[\mathrm{nA}], \quad \beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$$

U, Γ, material	E_J/h	I_c	$E_J/10h$	$E_J/10k_B$	$L_{\beta=1}$
2Δ, $\Delta/2$, Al	0.32 GHz	0.65 nA	32 MHz	1.5 mK	510 nH
2Δ, $\Delta/2$, NbTiN	2.1 GHz	4.3 nA	210 MHz	10 mK	76 nH
6Δ, 2Δ, NbTiN	9.9 GHz	20 nA	0.99 GHz	47 mK	17 nH
6Δ, 2Δ, Nb	11.5 GHz	23 nA	1.15 GHz	55 mK	14 nH

- Condensate and cat need $E_J/\hbar\omega \gtrsim 5$ and $\hbar\omega > k_B T$: a strong dot junction in NbTiN/Nb with a $\simeq 1\,\mathrm{GHz}$ mode, $L \simeq 17\,\mathrm{nH}$, $Z \simeq 160\,\Omega$ (full DMRG: 9.6 photons, $0.94 I_c$).
- The flux-driven 0–π switch needs only $\hbar\omega = 0.03$–0.3Δ and $Z = 0.7$–$3\,\mathrm{k}\Omega$: Al, 1–13 GHz, granular-Al superinductors.

Backup: how the current was computed

- Current operator $I = -(e/\hbar)\partial H/\partial\varphi_{\mathrm{cl}}$ through the dot bond, dressed by D; checked against the sum rule $\langle I\rangle = (e/\hbar)(\hbar\omega/2g)\langle a + a^{\dagger}\rangle$ at every point (median violation $10^{-4} I_c$).
- Seed $\varphi_{\mathrm{cl}} = 10^{-3}$ to select a direction; two independent DMRG starts (vacuum side / polarized side) with seed annealing, because the two current states are split only by δ and by $2I^{\star}\varphi_{\mathrm{cl}} \sim 10^{-5}\Delta$.
- Seed 10^{-4}: current unchanged above threshold, reduced $\times 0.09$ below – it is spontaneous.
- $\chi = 400$ changes energies by $< 4 \times 10^{-8}\Delta$ and the current by $< 0.3\%$ away from the crossover.